

Project Kaushal Setu

A unified government digital infrastructure connecting certified youth with local industry — privacy-first, federated, and built for scale.

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The Invisible Grassroots Worker Crisis

Millions of certified skilled workers remain effectively invisible to local employers because training centres, NGOs and CSR teams hold isolated records with no live interoperability.



Worker barriers

Low digital literacy, no standard identity, and limited access to national job portals leave certified workers underemployed.



Employer pain

MSMEs cannot reliably discover local, verified talent — searches are slow and verification is manual.

Fragmented systems

Institutes retain their own databases; there is no secure federation or consent-driven sharing mechanism.

Real-life example: A certified welder from Dharavi registered with a local NGO remains invisible to a construction company 5 km away in Andheri — due to disconnected records and manual processes.

Project Kaushal Setu — Proposed Solution

India's Digital Public Infrastructure for Skilled Employment



No Data Migration

Institutes run their own databases; Kaushal Setu acts as a federated query layer using REST APIs.



Shramik ID

Blockchain-verified digital identity linked to Aadhaar, issued by training institutes.



Federated Search & AI Match

Real-time cross-database search with AI ranking by skill match (85%+ accuracy), distance (5-50km radius), language preference, and availability.



WhatsApp Consent

Workers receive SMS/WhatsApp in regional language; reply '1' to share profile; audit trail maintained.



Plug-and-Play API

15-minute integration with NVEQF, NASSCOM, and state skill mission databases.

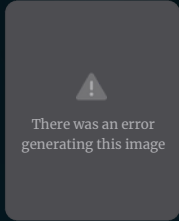
How it works: Verify → Discover → Consent → Hire. Student completes NVEQF course → Institute issues Shramik ID → Employer searches locally → AI ranks matches → Worker consents via WhatsApp → Verified hire. Zero data leaves institute servers without consent.

How Kaushal Setu Works — System Workflow



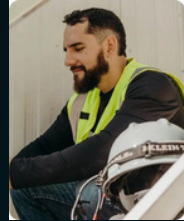
This consent-driven flow ensures verification, discoverability and traceable hiring while preserving institute ownership and worker privacy.

The System in Action — Case Study



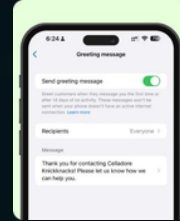
Amit's Journey

Amit completes a six-month electrician course at a government vocational centre; his institute issues a verified record and Shramik ID.



Discovery & Match

A local builder searches for electricians within 5 km. AI recommends Amit based on skills, proximity and availability.



Consent & Connect

Amit receives a Marathi WhatsApp message, replies '1' to consent; the builder instantly receives Amit's verified contact and certificates.



Hiring & Updates

Interview scheduled, hiring completed, and employment status is updated across participating databases to improve future matching and analytics.

Why feasible: uses existing systems, lightweight API, consent-first flows, regional language support and minimal behavioural change for institutes and workers.

Kaushal Setu: Bridging India's Skill Gap

Verified talent. Faster hiring. Stronger local opportunity.

Conclusion

- Verified profiles build trust
- Consent-first flow protects workers
- AI matching speeds up hiring
- Local jobs become easier to find
- Institutes gain a direct outcome loop

Future Scope

- Expand across more trades and regions
- Integrate deeper with MIS and CSR portals
- Add stronger analytics for demand sensing
- Support more Indian languages
- Scale to broader public and private hiring networks

From invisible workers to verified talent. From manual hiring to AI-powered discovery. Kaushal Setu is India's answer to grassroots employment.